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Unsupervised Self-Normalized Change-Point Testing for Time Series

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Abstract

We propose a new self-normalized method for testing change points in the time series setting. Self-normalization has been celebrated for its ability to avoid direct estimation of the nuisance asymptotic variance and its flexibility of being generalized to handle quantities other than the mean. However, it was developed and mainly studied for constructing confidence intervals for quantities associated with a stationary time series, and its adaptation to change-point testing can be nontrivial as direct implementation can lead to tests with nonmonotonic power. Compared with existing results on using self-normalization in this direction, the current paper proposes a new self-normalized change-point test that does not require prespecifying the number of total change points and is thus unsupervised. In addition, we propose a new contrast-based approach in generalizing self-normalized statistics to handle quantities other than the mean, which is specifically tailored for change-point testing. Monte Carlo simulations are presented to illustrate the finite-sample performance of the proposed method.

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1 Introduction

The problem of assessing structural stability has been an important topic in various fields including climate science, economics, finance and industrial engineering among others. It has been traditionally phrased as hypothesis testing problems with the null hypothesis indicating structural stability and the alternative containing one or multiple structural breaks, and one often relies heavily on statistical reasoning to find appropriate solutions. As commented in a recent review paper by Aue and Horváth (2013), much of the methodology was first developed for independent observations, while in many applications involving time series data it is of interest to incorporate serial dependence into the analysis for which two approaches have emerged. The first one is to impose certain parametric assumptions to explicitly characterize the dependence structure and potential structural breaks, and as a result likelihood methods are often used to devise the corresponding test statistics. Nevertheless, such an approach can be vulnerable to model misspecification, in which case one chooses to seek an alternative, namely the second approach, that does not require the explicit specification of a certain time series model by instead studying the effect of dependence on the asymptotic distribution of given test statistics. To correct for serial dependence, it often requires a consistent estimator of the long-run variance, namely the spectral density at zero frequency. The later quantity involves autocovariances of all orders, and a data-driven bandwidth is typically needed for its estimator to be adaptive to the underlying dependence. However, existing results on long-run variance estimation and its bandwidth selection were usually developed under structural stability, and thus can be biased under the change-point alternative leading to tests with nonmonotonic power, namely the power of the test can decrease in some ranges as the alternative deviates from the null; see Vogelsang (1999) and Shao and Zhang (2010) for further discussions.

To avoid the unpleasant issue of nonmonotonic power, Shao and Zhang (2010) considered adopting the self-normalization idea of Lobato (2001) and Shao (2010b), and extending it to the change-point setting. In self-normalization, instead of resorting to a consistent estimator of the long-run variance, one relies on a sequence of recursive estimators to form the normalizer and in turn pivotalize the asymptotic distribution of the test statistic. We shall here refer the readers to the recent review paper by Shao (2015) for further discussions and references on self-normalization. Note that the idea of self-normalization was originally developed in the context of constructing confidence intervals for quantities associated with a stationary time series, and its adaptation to change-point testing can be nontrivial. In particular, Shao and Zhang (2010) demonstrated that direct implementation of self-normalization can lead to inconsistent change-point tests, namely

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the power of the resulting test may not necessarily increase to one as the alternative deviates from the null. To take the change-point alternative into account, the aforementioned paper proposed to consider the situation with a single change-point alternative and devised a self-normalized test that has a monotonic power function. Extensions to multiple change-point alternatives were also given in the same paper, but the form of the test statistic, and thus its detailed implementation, depend on the number of actual change points, which however is typically unknown in practice.

In the current paper, we propose a new self-normalized test that does not require the *a priori* information on the number of actual change points yet still leads to a valid statistical procedure for testing change points. This is considered to be important because (i) the actual number of change points as required by the test of Shao and Zhang (2010) is typically unknown in practice; and (ii) when the number of change points is misspecified, the test of Shao and Zhang (2010) can suffer from power loss and in some situations the power can decrease to zero as the alternative deviates from the null; see the simulation study in Section 4.2. Shao and Zhang (2010) suggested estimating the number of change points before calling their testing procedure, for which they mentioned several candidates including the information criterion of Yao (1988), the minimum description length criterion of Davis et al. (2006), and the sequential testing procedure of Bai and Perron (1998). Since different methods may yield different estimates for the number of change points, selecting the optimal one for this preliminary estimation step can be viewed as a tuning parameter selection problem for the test of Shao and Zhang (2010), and it can be a nontrivial task as the candidates mentioned by Shao and Zhang (2010) were mostly developed in different settings. For example, the information criterion of Yao (1988) was developed for independent Gaussian observations (see also Zhang and Siegmund, 2007 for an extension to comparative genomic hybridization data), while the method of Davis et al. (2006) is specifically tailored to autoregressive models. In addition, from a practical point of view, it would be philosophically more natural to first consider a global test for the existence of change points, and if the existence is confirmed by the test, then further analysis may be called by the investigator to estimate the actual number and locations of change points. Therefore, it seems desirable if we can have a universal test as proposed in the current paper that does not require the *a priori* information on the number of actual change points. We also mention that the test of Shao and Zhang (2010) may become computationally prohibitive when the number of change points exceeds two, as it relies on simulation to obtain quantiles of the asymptotic distribution which is nonstandard and involves a maximization over a space whose dimension grows exponentially with the number of change points. In contrast, the current test, and thus its computational burden, do not depend on the number of change points, and we also pro-

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pose an approximation scheme to further facilitate the computation needed in the proposed testing procedure.

Another improvement that the current paper aims to make is to propose a new approach in generalizing self-normalized statistics to handle quantities other than the mean. In the mean case, self-normalized statistics are generally functions of the partial sum process, which after suitable scaling represents a sequence of recursive mean estimators. Hence, when one is interested in quantities other than the mean, the conventional approach is then to replace the role that the partial sum process plays in the mean case by recursive estimators of the quantity of interest. Such an approach was originally advocated by Shao (2010b) in the context of constructing confidence intervals for quantities associated with a stationary time series. However, due to its naturalness and ease of implementation, it gradually becomes a standard and has been adopted into various settings including change-point testing (Shao and Zhang, 2010). With this conventional approach, change-point testing is then conducted by sequentially comparing the recursive estimate with the overall estimate. Under the alternative, the overall estimate is computed by using observations both before and after the change point, and is thus expected to be different from the recursive estimate that is computed by using observations only before the hypothesized change point. However, when one is interested in robust quantities like the median, the overall estimate in this case can be more resistant to the inhomogeneity caused by the change-point alternative, making the power of the resulting test to be suboptimal. To overcome the problem, we propose a contrast-based approach for generalizing self-normalized statistics to handle quantities other than the mean, which is specifically tailored for change-point testing problems. The consistency of the resulting change-point test is also established for cases with both the mean and quantities beyond the mean. Note that Shao and Zhang (2010) only studied the consistency of their test in the mean case.

The rest of the paper is organized as follows. Section 2 considers the case of the mean, and proposes a new self-normalized statistic for unsupervised change-point testing. Its generalizations to cases beyond the mean are provided in Section 3, using the conventional approach in Section 3.1 and the proposed new contrast-based approach in Section 3.2. Asymptotic properties including the limiting null distribution and power behavior are studied in the respective sections. Numerical experiments including Monte Carlo simulations and a real data analysis are conducted in Section 4 to illustrate the finite-sample performance. Technical proofs are deferred to the Appendix.

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2 Testing for Change Points in the Mean

To illustrate the idea, we first consider the case for the mean. Given an observed time series $X_1, \ldots, X_n$, the problem is to test the null hypothesis of no change point in the mean, namely

$$H_0 : \; E(X_1) = \cdots = E(X_n) = \mu, \tag{1}$$

versus the alternative of having one or multiple change points, namely there exist $M \geq 1$ break points $k_0 = 1 < k_1 < \cdots < k_M < n = k_{M+1}$ such that

$$E(X_i) \neq E(X_{i+1}), \quad i \in \{k_1, \ldots, k_M\}, \tag{2}$$

and $E(X_i) = E(X_{i+1})$ otherwise. The actual number of change points M and their specific locations $k_1, \ldots, k_M$ are typically unknown in practice, especially when one is at the stage of seeking a global test for determining their existence. Let $\bar{X}_{j,k} = (k-j+1)^{-1} \sum_{i=j}^{k} X_i$ be the sample mean of $X_j, \ldots, X_k$ and $S_{j,k} = \sum_{i=j}^{k} X_i$ be the associated partial sum, $1 \leq j \leq k \leq n$. To devise a statistical test for (1), a commonly adopted strategy is to compare the recursive mean $\bar{X}_{1,k} = S_{1,k}/k$ with the global mean $\bar{X}_{1,n} = S_{1,n}/n$ and track if there is any significant deviation between the two, which leads to the cumulative sum (CUSUM) process

$$Z_n(t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} (X_i - \bar{X}_{1,n}), \quad t \in [0, 1],$$

where $\lfloor nt \rfloor$ is the largest integer not exceeding nt. Under the null hypothesis of no change point, the process $\{Z_n(t)\}_{t \in [0,1]}$ is free of the unknown but constant mean parameter μ, and obtaining the appropriate cut-off value requires an asymptotic theory on the (centered) partial sum process

$$S_n(t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} (X_i - \mu).$$

Let $\boldsymbol{B}(\cdot)$ be a standard Brownian motion and $\Rightarrow$ denote weak convergence in the Skorokhod space, we shall here make the following assumption.

(IP) There exists a $\sigma > 0$ such that $\{S_n(t), 0 \leq t \leq 1\} \Rightarrow \{\sigma \boldsymbol{B}(t), 0 \leq t \leq 1\}$.

Assumption (IP) is generally referred to as the invariance principle, and has been proven to hold for a wide range of short-range dependent processes with the help of various dependence measures; see for example the mixing coefficients of Hannan (1979) and Herrndorf (1984), the functional

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dependence measure of Wu (2007) and Berkes et al. (2014), and references therein for other contributions. By Assumption (IP) and the continuous mapping theorem, one can obtain the weak convergence that

$$Z_n(t) \to_D \sigma\{\boldsymbol{B}(t) - t\boldsymbol{B}(1)\},$$

where the asymptotic distribution depends on the nuisance parameter σ. For time series data, the quantity σ^2 is typically the long-run variance which involves autocovariances of all orders, and a data-driven bandwidth is often needed for its estimator to be adaptive to the underlying dependence strength. However, as commented by Shao and Zhang (2010), both theoretical and empirical studies in the literature have found that the use of data-dependent bandwidth can lead to tests with nonmonotonic power. To alleviate the problem, Shao and Zhang (2010) proposed to adopt the idea of self-normalization (Lobato, 2001 and Shao, 2010b) and generalize it to the change-point setting. The main ingredient of self-normalization is to consider using a function of the CUSUM process as the normalizer to pivotalize the asymptotic distribution. In particular, let

$$V_n^2 = \frac{1}{n^2}\sum_{i=1}^{n}\left\{\sum_{j=1}^{i}(X_j - \bar{X}_{1,n})\right\}^2,$$

then the continuous mapping theorem implies that

$$\frac{Z_n(t)^2}{V_n^2} \to_D \frac{\{\boldsymbol{B}(t) - t\boldsymbol{B}(1)\}^2}{\int_0^1\{\boldsymbol{B}(s) - s\boldsymbol{B}(1)\}^2 ds},$$

where the asymptotic distribution no longer depends on the nuisance parameter σ and can thus be used to obtain the cut-off values. Nevertheless, this simple application of self-normalization unfortunately does not yield good performance for change-point testing, and the main reason is that both $Z_n(t)$ and V_n can be affected by the change-point alternative, making the resulting test of less or no power; see also the simulation demonstration of Shao and Zhang (2010). To overcome the problem, Shao and Zhang (2010) proposed to split the data for calculating the self-normalizer according to the partition implied by hypothesized change-point locations (whose total number is assumed to be known), and after forming the self-normalized statistics for each possible partition with a prespecified number of segments, the maximum is then taken as the final test statistic. Such an approach is the most ideal for the situation with a single change-point alternative, on which the paper of Shao and Zhang (2010) was mainly focused. However, its application to situations with possibly more than one change points can pose certain challenges. To be more specific,

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the test of Shao and Zhang (2010) was developed under a supervised learning setting where the total number of change points needs to be prespecified, which however is typically unknown in practice especially when one is at the preliminary stage of seeking a statistical test to determine the existence of change points. In addition, as the test of Shao and Zhang (2010) requires the calculation of self-normalized statistics for all possible partitions with a prespecified number of segments, its computational burden will grow exponentially with the prespecified number of total change points. This, together with the large simulation needed to approximate the nonstandard asymptotic distribution with the same computational mechanism, makes the test of Shao and Zhang (2010) computationally prohibitive (or at least less favorable) for situations with more than two change points. Moreover, our simulation study in Section 4.2 indicates that, when the number of change points is misspecified, the test of Shao and Zhang (2010) can suffer from power loss and in some situations the power can decrease to zero as the alternative deviates from the null.

We shall here propose an alternative that does not require the *a priori* information on the number of change points yet still leads to a valid statistical procedure for change-point testing. The computational burden of the proposed test will remain at the same for situations with different numbers of change points. We shall here introduce the following notations:

$$D_{n,f}(j_1, j_2, j_3) = \frac{j_2 - j_1 + 1}{\sqrt{j_3 - j_1 + 1}}(\bar{X}_{j_1,j_2} - \bar{X}_{j_1,j_3}); \tag{3}$$

$$L_{n,f}(j_1, j_2, j_3) = \sum_{i=j_1}^{j_2} \left(\frac{i - j_1 + 1}{j_3 - j_1 + 1}\right)^2 (\bar{X}_{j_1,i} - \bar{X}_{j_1,j_2})^2; \tag{4}$$

$$R_{n,f}(j_1, j_2, j_3) = \sum_{i=j_2+1}^{j_3} \left(\frac{j_3 - i + 1}{j_3 - j_1 + 1}\right)^2 (\bar{X}_{i,j_3} - \bar{X}_{j_2+1,j_3})^2. \tag{5}$$

Since (3) has the equivalent form $D_{n,f}(j_1, j_2, j_3) = (j_3 - j_1 + 1)^{-1/2} \sum_{i=j_1}^{j_2}(X_i - \bar{X}_{j_1,j_3})$, it relates to the CUSUM process associated with $X_{j_1}, \ldots, X_{j_3}$ and recursively compares the forwarding partial mean $\bar{X}_{j_1,j_2}$, $j_2 \leq j_3$, with the overall mean $\bar{X}_{j_1,j_3}$ to seek for potential change points. To make the normalization process not affected by the change-point alternative, one-sided self-normalizers are

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used and are given in (4) and (5) respectively. Similarly, we introduce the backward version:

$$
\begin{aligned}
D_{n,b}(j_1, j_2, j_3) &= \frac{j_3 - j_2 + 1}{\sqrt{j_3 - j_1 + 1}}(\bar{X}_{j_2,j_3} - \bar{X}_{j_1,j_3}); \\
L_{n,b}(j_1, j_2, j_3) &= \sum_{i=j_1}^{j_2-1} \left(\frac{i - j_1 + 1}{j_3 - j_1 + 1}\right)^2 (\bar{X}_{j_1,i} - \bar{X}_{j_1,j_2-1})^2; \\
R_{n,b}(j_1, j_2, j_3) &= \sum_{i=j_2}^{j_3} \left(\frac{j_3 - i + 1}{j_3 - j_1 + 1}\right)^2 (\bar{X}_{i,j_3} - \bar{X}_{j_2,j_3})^2.
\end{aligned}
$$

Let $\Omega(\varepsilon) = \{(t_1, t_2) : \ \varepsilon \le t_1 < t_2 \le 1 - \varepsilon, \ t_2 - t_1 \ge \varepsilon\}$, $\Omega_n(\varepsilon) = \{(\lfloor nt_1 \rfloor, \lfloor nt_2 \rfloor) : \ (t_1, t_2) \in \Omega(\varepsilon)\}$, $\Xi_{n,f}(j_1, j_2, j_3) = L_{n,f}(j_1, j_2, j_3) + R_{n,f}(j_1, j_2, j_3)$ and $\Xi_{n,b}(j_1, j_2, j_3) = L_{n,b}(j_1, j_2, j_3) + R_{n,b}(j_1, j_2, j_3)$, our test statistic is then defined as

$$
T_n = \max_{(l_1,l_2)\in\Omega_n(\varepsilon)} \frac{D_{n,f}(1, l_1, l_2)^2}{\Xi_{n,f}(1, l_1, l_2)} + \max_{(m_1,m_2)\in\Omega_n(\varepsilon)} \frac{D_{n,b}(m_1, m_2, n)^2}{\Xi_{n,b}(m_1, m_2, n)}. \tag{6}
$$

Unlike the supervised test of Shao and Zhang (2010) which detects change points by formulating piecewise stationary partitions according to the prespecified number of change points, the proposed test is unsupervised and detects change points by recursive scanning. The following theorem provides the asymptotic property of T_n under both the null and alternative.

Theorem 2.1. *Assume condition (IP). Then (i) under the null hypothesis (1), we have*

$$
T_n \to_D T(\boldsymbol{B}) := \sup_{(r_1,r_2)\in\Omega(\varepsilon)} \frac{D(\boldsymbol{B}, 0, r_1, r_2)^2}{\Xi(\boldsymbol{B}, 0, r_1, r_2)} + \sup_{(s_1,s_2)\in\Omega(\varepsilon)} \frac{D(\boldsymbol{B}, s_1, s_2, 1)^2}{\Xi(\boldsymbol{B}, s_1, s_2, 1)}, \tag{7}
$$

where

$$
\begin{aligned}
D(\boldsymbol{B}, t_1, t_2, t_3) &= \frac{1}{\sqrt{t_3 - t_1}} \left[\boldsymbol{B}(t_2) - \boldsymbol{B}(t_1) - \frac{t_2 - t_1}{t_3 - t_1}\{\boldsymbol{B}(t_3) - \boldsymbol{B}(t_1)\}\right], \\
\Xi(\boldsymbol{B}, t_1, t_2, t_3) &= \frac{1}{(t_3 - t_1)^2} \left(\int_{t_1}^{t_2} \left[\boldsymbol{B}(s) - \boldsymbol{B}(t_1) - \frac{s - t_1}{t_2 - t_1}\{\boldsymbol{B}(t_2) - \boldsymbol{B}(t_1)\}\right]^2 ds \right. \\
&\quad \left. + \int_{t_2}^{t_3} \left[\boldsymbol{B}(t_3) - \boldsymbol{B}(s) - \frac{t_3 - s}{t_3 - t_2}\{\boldsymbol{B}(t_3) - \boldsymbol{B}(t_2)\}\right]^2 ds \right);
\end{aligned}
$$

and (ii) under the alternative (2) with $\min_{0\le i\le M} |k_{i+1} - k_i|/n > \varepsilon$ *and* $E(X_{k_i+1}) - E(X_{k_i}) = n^{-1/2} L_i$ *for some* $L_i \neq 0$, $i = 1, \ldots, M$, *we have* $T_n \to \infty$ *in probability if* $\min_{1\le i\le M} |L_i| \to \infty$.

By (7), we reject the null hypothesis (1) at level $\alpha \in (0, 1)$ if $T_n > q_{1-\alpha}$ where $q_{1-\alpha}$ is the $(1-\alpha)$-th quantile of $T(\boldsymbol{B})$, and its consistency is guaranteed by Theorem 2.1 (ii). The above testing

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procedure involves a trimming parameter ε, which controls the minimal length (in proportion) of partial sums. As commented by Zhou and Shao (2013), trimming parameters in self-normalization are different from smoothing parameters, such as the truncation lag in long-run variance estimation (Newey and West, 1987, Andrews, 1991, Liu and Wu, 2010, Politis, 2011), the window size in dependent bootstrap (Künsch, 1989, Lahiri, 2003, Shao, 2010a, Zhou, 2013) or the subsampling width in subsampling methods (Hall and Jing, 1996, Politis et al., 1999, Zhang et al., 2013), as the effect of trimming is accounted for in the limiting distribution and its approximation. Zhou and Shao (2013), along with Huang et al. (2015), conducted a series of numerical analyses and argued that the rule of thumb choice of $\varepsilon = 0.1$ seems to yield satisfactory performance for self-normalized methods. The proposed test has the potential of being generalized to quantities other than the mean, for example to quantiles, which we shall discuss in Section 3.

3 Extensions to Other Quantities of Interest

3.1 The Conventional Approach

An attractive feature of self-normalization is that the resulting method can be readily generalized from the mean case to other quantities of interest, for example to quantiles. The key idea of achieving this is to exploit the role that the CUSUM process plays in the mean case, and replace it by an appropriate substitute in the general setting. To be more specific, the CUSUM process

$$\begin{aligned} Z_n(t) &= \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} (X_i - \bar{X}_n) \\ &= \frac{1}{\sqrt{n}} \left(\sum_{i=1}^{\lfloor nt \rfloor} X_i - \lfloor nt \rfloor \bar{X}_n \right) = \frac{\lfloor nt \rfloor}{\sqrt{n}} (\bar{X}_{\lfloor nt \rfloor} - \bar{X}_n), \quad t \in [0,1], \end{aligned} \tag{8}$$

represents the difference between the recursive mean estimator $\bar{X}_{\lfloor nt \rfloor}$ and the overall mean estimator $\bar{X}_n$. Hence, if one is interested in a general parameter $\boldsymbol{\theta}$ that is possibly different from the mean, then it would be natural to consider replacing the $\bar{X}_{\lfloor nt \rfloor}$ and $\bar{X}_n$ in (8) by the corresponding recursive estimator $\hat{\boldsymbol{\theta}}_{\lfloor nt \rfloor}$ and $\hat{\boldsymbol{\theta}}_n$, making the new statistic a function of the process $(\lfloor nt \rfloor / \sqrt{n})(\hat{\boldsymbol{\theta}}_{\lfloor nt \rfloor} - \hat{\boldsymbol{\theta}}_n)$, $t \in [0,1]$. Such an approach was advocated by Shao (2010b) in the context of confidence interval construction, and gradually becomes a standard in generalizing self-normalized methods to quantities other than the mean; see for example the application in Shao and Zhang (2010) for change-point testing and Shao (2015) for a recent review. We shall here consider a generalization

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of the test proposed in Section 2 by following this conventional approach.

To facilitate the mathematical formulation, suppose one is interested in the quantity $\boldsymbol{\theta}_i = \boldsymbol{Q}(\boldsymbol{F}_i) \in \mathbb{R}^d$, a function of $\boldsymbol{F}_i$, the distribution of $\boldsymbol{Y}_i = (X_i, X_{i+1}, \ldots, X_{i+\ell-1})$. Then we consider the problem of testing for the null hypothesis

$$H_0^\star : \boldsymbol{\theta}_1 = \cdots = \boldsymbol{\theta}_N = \boldsymbol{\vartheta}, \quad N = n - \ell + 1, \tag{9}$$

versus the alternative that there exist $M^\star \geq 1$ break points $k_0^\star = 1 < k_1^\star < \cdots < k_{M^\star}^\star < N = k_{M^\star+1}^\star$ such that

$$\boldsymbol{\theta}_i \neq \boldsymbol{\theta}_{i+1}, \quad i \in \{k_1^\star, \ldots, k_M^\star\}, \tag{10}$$

and $\boldsymbol{\theta}_i = \boldsymbol{\theta}_{i+1}$ otherwise. Let $\hat{\boldsymbol{F}}_{j,k}$ be the empirical distribution of $(\boldsymbol{Y}_j, \ldots, \boldsymbol{Y}_k)$, and denote by $\boldsymbol{F}_{j,k}$ the uniform mixture of $\boldsymbol{F}_j, \ldots, \boldsymbol{F}_k$, $1 \leq j \leq k \leq N$. Let $\hat{\boldsymbol{\theta}}_{j,k} = \boldsymbol{Q}(\hat{\boldsymbol{F}}_{j,k})$, then counterparts of (3)–(5) are given by

$$\boldsymbol{D}_{N,f}^\star(j_1, j_2, j_3) = \frac{j_2 - j_1 + 1}{\sqrt{j_3 - j_1 + 1}}(\hat{\boldsymbol{\theta}}_{j_1,j_2} - \hat{\boldsymbol{\theta}}_{j_1,j_3}), \tag{11}$$

$$\boldsymbol{L}_{N,f}^\star(j_1, j_2, j_3) = \sum_{i=j_1}^{j_2} \left(\frac{i - j_1 + 1}{j_3 - j_1 + 1}\right)^2 (\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{j_1,j_2})(\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{j_1,j_2})^\top, \tag{12}$$

$$\boldsymbol{R}_{N,f}^\star(j_1, j_2, j_3) = \sum_{i=j_2+1}^{j_3} \left(\frac{j_3 - i + 1}{j_3 - j_1 + 1}\right)^2 (\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2+1,j_3})(\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2+1,j_3})^\top. \tag{13}$$

Similarly, let

$$\boldsymbol{D}_{N,b}^\star(j_1, j_2, j_3) = \frac{j_3 - j_2 + 1}{\sqrt{j_3 - j_1 + 1}}(\hat{\boldsymbol{\theta}}_{j_2,j_3} - \hat{\boldsymbol{\theta}}_{j_1,j_3}),$$

$$\boldsymbol{L}_{N,b}^\star(j_1, j_2, j_3) = \sum_{i=j_1}^{j_2-1} \left(\frac{i - j_1 + 1}{j_3 - j_1 + 1}\right)^2 (\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{j_1,j_2-1})(\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{j_1,j_2-1})^\top,$$

$$\boldsymbol{R}_{N,b}^\star(j_1, j_2, j_3) = \sum_{i=j_2}^{j_3} \left(\frac{j_3 - i + 1}{j_3 - j_1 + 1}\right)^2 (\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2,j_3})(\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2,j_3})^\top,$$

$\boldsymbol{\Xi}_{N,f}^\star(j_1, j_2, j_3) = \boldsymbol{L}_{N,f}^\star(j_1, j_2, j_3) + \boldsymbol{R}_{N,f}^\star(j_1, j_2, j_3)$ and $\boldsymbol{\Xi}_{N,b}^\star(j_1, j_2, j_3) = \boldsymbol{L}_{N,b}^\star(j_1, j_2, j_3) + \boldsymbol{R}_{N,b}^\star(j_1, j_2, j_3)$, then the generalized test statistic can be defined as

$$\begin{aligned} T_N^\star = & \max_{(l_1,l_2)\in\Omega_N(\varepsilon)} \boldsymbol{D}_{N,f}^\star(1, l_1, l_2)^\top \boldsymbol{\Xi}_{N,f}^\star(1, l_1, l_2)^{-1} \boldsymbol{D}_{N,f}^\star(1, l_1, l_2) \\ & + \max_{(m_1,m_2)\in\Omega_N(\varepsilon)} \boldsymbol{D}_{N,b}^\star(m_1, m_2, N)^\top \boldsymbol{\Xi}_{N,b}^\star(m_1, m_2, N)^{-1} \boldsymbol{D}_{N,b}^\star(m_1, m_2, N). \end{aligned} \tag{14}$$

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Let $\boldsymbol{B}(\cdot)$ be a d-dimensional standard Brownian motion and $\mathcal{D} = \{(t_1, t_2) : \ 0 \le t_1 \le t_2 \le 1\}$, we shall here make the following assumption on the recursive estimators $\hat{\boldsymbol{\theta}}_{j,k} = \boldsymbol{Q}(\hat{\boldsymbol{F}}_{j,k})$.

(IP*) There exists a positive definite matrix $\boldsymbol{\Sigma}$ such that

$$\left[\frac{\lfloor Nt\rfloor - \lfloor Ns\rfloor \vee 1 + 1}{\sqrt{N}}\{\boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor Ns\rfloor\vee 1,\lfloor Nt\rfloor}) - \boldsymbol{Q}(\boldsymbol{F}_{\lfloor Ns\rfloor\vee 1,\lfloor Nt\rfloor})\}, (s,t) \in \mathcal{D}\right] \rightsquigarrow [\boldsymbol{\Sigma}\{\boldsymbol{B}(t) - \boldsymbol{B}(s)\}, (s,t) \in \mathcal{D}],$$

where $\lfloor Ns\rfloor \vee 1 = \max(\lfloor Ns\rfloor, 1)$ and $\rightsquigarrow$ denotes the weak convergence in the sense of Hoffmann-Jørgensen (van der Vaart and Wellner, 1996).

Note that the partial sum process in Assumption (IP) relates to the empirical process, Assumption (IP*) provides a counterpart in the current general setting with the additional transform $\boldsymbol{Q}$, and can be verified by using the functional delta method of Volgushev and Shao (2014); see also the influence function approach of Shao (2010b). Let

$$\boldsymbol{\Delta}(\boldsymbol{B}, r, s, t, u) = \boldsymbol{B}(s) - \boldsymbol{B}(r) - \frac{s-r}{u-t}\{\boldsymbol{B}(u) - \boldsymbol{B}(t)\},$$

the following theorem provides an asymptotic theory on the test statistic $T_N^\star$.

Theorem 3.1. *Assume condition (IP*). Then (i) under the null hypothesis (9), we have*

$$\begin{aligned} T_N^\star \to_D T^\star(\boldsymbol{B}) \ := & \sup_{(r_1,r_2)\in\Omega(\varepsilon)} \boldsymbol{D}^\star(\boldsymbol{B}, 0, r_1, r_2)^\top \boldsymbol{\Xi}^\star(\boldsymbol{B}, 0, r_1, r_2)^{-1} \boldsymbol{D}^\star(\boldsymbol{B}, 0, r_1, r_2) \\ & + \sup_{(s_1,s_2)\in\Omega(\varepsilon)} \boldsymbol{D}^\star(\boldsymbol{B}, s_1, s_2, 1)^\top \boldsymbol{\Xi}^\star(\boldsymbol{B}, s_1, s_2, 1)^{-1} \boldsymbol{D}^\star(\boldsymbol{B}, s_1, s_2, 1), \end{aligned} \tag{15}$$

where

$$\begin{aligned} \boldsymbol{D}^\star(\boldsymbol{B}, t_1, t_2, t_3) &= \frac{1}{\sqrt{t_3 - t_1}} \boldsymbol{\Delta}(\boldsymbol{B}, t_1, t_2, t_1, t_3), \\ \boldsymbol{\Xi}^\star(\boldsymbol{B}, t_1, t_2, t_3) &= \frac{1}{(t_3 - t_1)^2}\left\{\int_{t_1}^{t_2} \boldsymbol{\Delta}(\boldsymbol{B}, t_1, s, t_1, t_2)\boldsymbol{\Delta}(\boldsymbol{B}, t_1, s, t_1, t_2)^\top ds \right. \\ & \left. + \int_{t_2}^{t_3} \boldsymbol{\Delta}(\boldsymbol{B}, s, t_3, t_2, t_3)\boldsymbol{\Delta}(\boldsymbol{B}, s, t_3, t_2, t_3)^\top ds\right\}; \end{aligned}$$

and (ii) under the alternative (10) with $\min_{0\le i\le M^\star} |k_{i+1}^\star - k_i^\star|/N > \varepsilon$*, we have* $T_N^\star \to \infty$ *in probability if* $\sqrt{N}|\hat{\boldsymbol{\theta}}_{k_{i-1}^\star+1,k_i^\star} - \hat{\boldsymbol{\theta}}_{k_{i-1}^\star+1,k_{i+1}^\star}| \to \infty$ *in probability,* $i = 1, \ldots, M^\star$.

Therefore, similar to the mean case, one rejects the null hypothesis (9) at level $\alpha \in (0, 1)$ if $T_N^\star > q_{1-\alpha}^\star$ where $q_{1-\alpha}^\star$ is the $(1-\alpha)$-th quantile of $T^\star(\boldsymbol{B})$ defined in (15). Nevertheless, in order to

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obtain the consistency of this test, unlike the mean case, one needs the stochastic condition that

$$\sqrt{N}|\hat{\boldsymbol{\theta}}_{k^\star_{i-1}+1,k^\star_i} - \hat{\boldsymbol{\theta}}_{k^\star_{i-1}+1,k^\star_{i+1}}| \to \infty \tag{16}$$

in probability as $n \to \infty$. Although Shao and Zhang (2010) did not study the power behavior of their test under this general setting, given the form of their test statistic and the argument used in the proof of Theorem 3.1, the same stochastic condition will also be required for their test to be consistent. Note that

$$\hat{\boldsymbol{\theta}}_{k^\star_{i-1}+1,k^\star_i} - \hat{\boldsymbol{\theta}}_{k^\star_{i-1}+1,k^\star_{i+1}} = \boldsymbol{Q}(\hat{\boldsymbol{F}}_{k^\star_{i-1}+1,k^\star_i}) - \boldsymbol{Q}(\hat{\boldsymbol{F}}_{k^\star_{i-1}+1,k^\star_{i+1}}),$$

where $\hat{\boldsymbol{F}}_{k^\star_{i-1}+1,k^\star_{i+1}}$ is the empirical distribution of $\boldsymbol{Y}_{k^\star_{i-1}+1}, \ldots, \boldsymbol{Y}_{k^\star_i}, \boldsymbol{Y}_{k^\star_i+1}, \ldots, \boldsymbol{Y}_{k^\star_{i+1}}$, a mixture of observations before and after the change point $k^\star_i$, which under the change-point alternative is considered to be different from $\hat{\boldsymbol{F}}_{k^\star_{i-1}+1,k^\star_i}$, the empirical distribution based only on observations before the change point. Hence, condition (16) seems to be a reasonable requirement for change-point detection. In fact, the following proposition states that, in the case of the mean, it can be reduced to its deterministic counterpart.

Proposition 3.1. *If $\theta_i = \int_{\mathbb{R}} x dF_i(x)$ represents the mean, then assuming (IP), condition (16) will be satisfied if $\sqrt{N}|E(X_{k^\star_i+1}) - E(X_{k^\star_i})| \to \infty$.*

However, for general nonlinear statistics, it is not always the case that the stochastic condition (16) required to guarantee the consistency as in Theorem 3.1 will be implied by its more desirable deterministic counterpart $\sqrt{N}|\boldsymbol{\theta}_{k^\star_i+1} - \boldsymbol{\theta}_{k^\star_i}| \to \infty$. To illustrate the idea, we consider a simple example in which $X_i = d\mathbf{1}_{\{i>pn\}}$, where $\mathbf{1}_{\{\cdot\}}$ is the indicator function and $p \in (0,1)$ represents the location (in proportion) of the change point. Hence, if one is interested in the mean, then the recursive mean estimate $\bar{X}_{\lfloor pn \rfloor} = 0$ and the overall mean estimate $\bar{X}_n = (1 - \lfloor pn \rfloor/n)d$, leading to the desirable result that $\sqrt{n}|\bar{X}_{\lfloor pn \rfloor} - \bar{X}_n| \to \infty$ if $\sqrt{n}d \to \infty$. However, if one is interested in nonlinear quantities like the median, then one can show that the recursive median estimate $\hat{\theta}_{\lfloor pn \rfloor} = 0$ and the overall median estimate $\hat{\theta}_n = 0$ when $p > 1/2 + 1/n$. Therefore, when the change point is located at the later half of the time series, the deterministic condition that $\sqrt{n}d \to \infty$ will no longer imply the needed stochastic condition that $\sqrt{n}|\hat{\theta}_{\lfloor pn \rfloor} - \hat{\theta}_n| \to \infty$, as $|\hat{\theta}_{\lfloor pn \rfloor} - \hat{\theta}_n| = 0$ in this case. Therefore, the CUSUM-type process $(\lfloor nt \rfloor/\sqrt{n})(\hat{\boldsymbol{\theta}}_{\lfloor nt \rfloor} - \hat{\boldsymbol{\theta}}_n)$, $t \in [0,1]$, may not be the most ideal for change-point detection in nonlinear quantities, especially when one is interested in robust quantities like the median that are potentially resistant to the inhomogeneity under the change-point alternative. This motivates us to seek for a new scheme for generalizing self-normalized test statistics to quantities

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other than the mean that is specifically tailored for change-point testing problems, which we shall consider in Section 3.2.

3.2 A New Approach: The Contrast-Based Method

By viewing $\bar{X}_{\lfloor nt \rfloor}$ as a recursive mean estimator, the conventional approach generalizes self-normalized statistics developed for the mean to more general quantities by replacing $(\bar{X}_{\lfloor nt \rfloor} - \bar{X}_n)$ in the CUSUM process with its recursive counterpart $(\hat{\boldsymbol{\theta}}_{\lfloor nt \rfloor} - \hat{\boldsymbol{\theta}}_n)$. However, as discussed in Section 3.1, it may not be the most ideal for change-point testing problems as it was originally developed for constructing confidence intervals by assuming structural stability. To provide a generalization that is directly related to change-point testing, we propose to focus on the contrast process $\{\boldsymbol{Q}(\hat{\boldsymbol{F}}_{1,\lfloor nt \rfloor}) - \boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor nt \rfloor+1,n})\}$, where $\boldsymbol{Q}(\hat{\boldsymbol{F}}_{1,\lfloor nt \rfloor})$ and $\boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor nt \rfloor+1,n})$ are parameter estimates based on $X_1, \ldots, X_{\lfloor nt \rfloor}$ and $X_{\lfloor nt \rfloor+1}, \ldots, X_n$ respectively. Hence, we are directly comparing estimates before and after the hypothesized change point, which is considered to be more relevant to change-point testing than the conventional $\{\boldsymbol{Q}(\hat{\boldsymbol{F}}_{1,\lfloor nt \rfloor}) - \boldsymbol{Q}(\hat{\boldsymbol{F}}_{1,n})\}$ for nonlinear quantities. This leads to the contrast-based counterparts of (11)–(13) given by

$$
\begin{aligned}
\boldsymbol{D}^{\diamond}_{N,f}(j_1, j_2, j_3) &= \frac{(j_2 - j_1 + 1)(j_3 - j_2)}{(j_3 - j_1 + 1)^{3/2}}(\hat{\boldsymbol{\theta}}_{j_1,j_2} - \hat{\boldsymbol{\theta}}_{j_2+1,j_3}), \\
\boldsymbol{L}^{\diamond}_{N,f}(j_1, j_2, j_3) &= \sum_{i=j_1}^{j_2} \frac{(i - j_1 + 1)^2 (j_2 - i)^2}{(j_3 - j_1 + 1)^2 (j_2 - j_1 + 1)^2}(\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{i+1,j_2})(\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{i+1,j_2})^\top, \\
\boldsymbol{R}^{\diamond}_{N,f}(j_1, j_2, j_3) &= \sum_{i=j_2+1}^{j_3} \frac{(i - 1 - j_2)^2 (j_3 - i + 1)^2}{(j_3 - j_1 + 1)^2 (j_3 - j_2)^2}(\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2+1,i-1})(\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2+1,i-1})^\top.
\end{aligned}
$$

Similarly, let

$$
\begin{aligned}
\boldsymbol{D}^{\diamond}_{N,b}(j_1, j_2, j_3) &= \frac{(j_2 - j_1)(j_3 - j_2 + 1)}{(j_3 - j_1 + 1)^{3/2}}(\hat{\boldsymbol{\theta}}_{j_2,j_3} - \hat{\boldsymbol{\theta}}_{j_1,j_2-1}), \\
\boldsymbol{L}^{\diamond}_{N,b}(j_1, j_2, j_3) &= \sum_{i=j_1}^{j_2-1} \frac{(i - j_1 + 1)^2 (j_2 - 1 - i)^2}{(j_3 - j_1 + 1)^2 (j_2 - j_1)^2}(\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{i+1,j_2-1})(\hat{\boldsymbol{\theta}}_{j_1,i} - \hat{\boldsymbol{\theta}}_{i+1,j_2-1})^\top, \\
\boldsymbol{R}^{\diamond}_{N,b}(j_1, j_2, j_3) &= \sum_{i=j_2}^{j_3} \frac{(i - j_2)^2 (j_3 - i + 1)^2}{(j_3 - j_1 + 1)^2 (j_3 - j_2 + 1)^2}(\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2,i-1})(\hat{\boldsymbol{\theta}}_{i,j_3} - \hat{\boldsymbol{\theta}}_{j_2,i-1})^\top,
\end{aligned}
$$

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$\boldsymbol{\Xi}^{\diamond}_{N,f}(j_1, j_2, j_3) = \boldsymbol{L}^{\diamond}_{N,f}(j_1, j_2, j_3) + \boldsymbol{R}^{\diamond}_{N,f}(j_1, j_2, j_3)$ and $\boldsymbol{\Xi}^{\diamond}_{N,b}(j_1, j_2, j_3) = \boldsymbol{L}^{\diamond}_{N,b}(j_1, j_2, j_3) + \boldsymbol{R}^{\diamond}_{N,b}(j_1, j_2, j_3)$, then our contrast-based generalization of (6) is given by

$$\begin{aligned} T^{\diamond}_N &= \max_{(l_1,l_2)\in\Omega_N(\varepsilon)} \boldsymbol{D}^{\diamond}_{N,f}(1, l_1, l_2)^{\top} \boldsymbol{\Xi}^{\diamond}_{N,f}(1, l_1, l_2)^{-1} \boldsymbol{D}^{\diamond}_{N,f}(1, l_1, l_2) \\ &\quad + \max_{(m_1,m_2)\in\Omega_N(\varepsilon)} \boldsymbol{D}^{\diamond}_{N,b}(m_1, m_2, N)^{\top} \boldsymbol{\Xi}^{\diamond}_{N,b}(m_1, m_2, N)^{-1} \boldsymbol{D}^{\diamond}_{N,b}(m_1, m_2, N). \end{aligned} \tag{17}$$

Note that in the case of the mean when $\theta_i = \int_{\mathbb{R}} x dF_i(x)$, we have $\hat{\theta}_{j,k} = S_{j,k}/(k - j + 1)$ and one can show that both the conventional generalization (14) and the contrast-based generalization (17) reduce to the same form, namely the test statistic (6) proposed in Section 2. However, test statistics in (14) and (17) are generally different when parameter estimators are not additive, namely when $\hat{\boldsymbol{\theta}}_{j_1,j_2} + \hat{\boldsymbol{\theta}}_{j_2+1,j_3} \neq \hat{\boldsymbol{\theta}}_{j_1,j_3}$, $j_1 \leq j_2 < j_3$. Unlike the generalization in (14) which relies on the conventional method that was originally developed for inference of constant parameters of a stationary time series, the proposed contrast-based test statistic (17) is specifically tailored for change-point testing and can be more powerful in detecting change points in robust quantities like the median. The following theorem provides its asymptotic distribution and theoretical consistency.

Theorem 3.2. *Assume condition (IP*). Then (i) under the null hypothesis (9), we have*

$$T^{\diamond}_N \to_D T^{\star}(\boldsymbol{B}),$$

where $T^{\star}(\boldsymbol{B})$ is given in (15); and (ii) under the alternative (10) with $\min_{0\leq i\leq M^\star} |k^\star_{i+1} - k^\star_i|/N > \varepsilon$ and $\boldsymbol{\theta}_{k^\star_i+1} - \boldsymbol{\theta}_{k^\star_i} = N^{-1/2} L^\star_i$ for some $L^\star_i \neq 0$, $i = 1, \ldots, M^\star$, we have $T^{\diamond}_N \to \infty$ in probability if $\min_{1\leq i\leq M^\star} |L^\star_i| \to \infty$.

Compared with Theorem 3.1, the contrast-based method no longer requires the stochastic condition (16). By Theorem 3.2 (ii), the proposed test will have unit asymptotic power if the magnitude of change is parametrically distinguishable, namely when the more desirable deterministic condition $N^{1/2}|\boldsymbol{\theta}_{k^\star_i+1} - \boldsymbol{\theta}_{k^\star_i}| \to \infty$ is satisfied.

4 Implementation

4.1 An Approximation Scheme

Note that the test statistic (14) involves a maximum of $O(N^2)$ terms, and its specific form does not depend on $M^\star$, the total number of change points. Therefore, the proposed testing procedure

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and its computational burden would be the same for situations with different numbers of change points. As a comparison, the test of Shao and Zhang (2010) requires the prespecification of the total number of change points $M^\star$, and their test involves the search of the maximum over $O(N^{M^\star})$ partitions. Therefore, besides the improvements of (i) not requiring the *a priori* knowledge on $M^\star$ (which is typically unknown in practice); and (ii) being more specifically tailored for change-point testing by using the contrast-based approach in handling quantities other than the mean, the proposed test is also considered to be computationally more advantageous when there are more than two change points, in which case the computational burden of the test of Shao and Zhang (2010) would grow exponentially with respect to their hypothesized or prespecified value of $M^\star$. We shall here in addition consider a grid approximation scheme to further reduce the computational burden of the current test, with which the maximum only needs to be searched over $O(N)$, instead of $O(N^2)$, partitions. To be more specific, let $\mathcal{G}_\varepsilon = \{(1+k\varepsilon)/2, k \in \mathbb{Z}\} \cap [0,1]$ be a grid on the unit interval, which is symmetric around the middle point. Let $\mathcal{G}_{\varepsilon,N,f} = \{(\lfloor Nt_1 \rfloor, \lfloor Nt_2 \rfloor) : \ (t_1,t_2) \in ([0,1] \times \mathcal{G}_\varepsilon) \cap \Omega(\varepsilon)\}$ and $\mathcal{G}_{\varepsilon,N,b} = \{(\lfloor Nt_1 \rfloor, \lfloor Nt_2 \rfloor) : \ (t_1,t_2) \in (\mathcal{G}_\varepsilon \times [0,1]) \cap \Omega(\varepsilon)\}$, we consider the test statistic

$$\begin{aligned} G_N^\diamond \ &= \ \max_{(l_1,l_2)\in\mathcal{G}_{\varepsilon,N,f}} \boldsymbol{D}_{N,f}^\diamond(1,l_1,l_2)^\top \boldsymbol{\Xi}_{N,f}^\diamond(1,l_1,l_2)^{-1} \boldsymbol{D}_{N,f}^\diamond(1,l_1,l_2) \\ &\quad + \max_{(m_1,m_2)\in\mathcal{G}_{\varepsilon,N,b}} \boldsymbol{D}_{N,b}^\diamond(m_1,m_2,N)^\top \boldsymbol{\Xi}_{N,b}^\diamond(m_1,m_2,N)^{-1} \boldsymbol{D}_{N,b}^\diamond(m_1,m_2,N), \end{aligned}$$

which provides a grid counterpart of the statistic $T_n^\diamond$ introduced in (17). The asymptotic distribution of $G_n^\diamond$ is given in Proposition 4.1 along with its consistency for testing the null hypothesis (9) against the change-point alternative.

Proposition 4.1. *Assume condition (IP*). Then (i) under the null hypothesis (9), we have*

$$G_N^\diamond \to_D G^\diamond(\boldsymbol{B}),$$

where

$$\begin{aligned} G^\diamond(\boldsymbol{B}) \ &= \ \sup_{(r_1,r_2)\in([0,1]\times\mathcal{G}_\varepsilon)\cap\Omega(\varepsilon)} \boldsymbol{D}^\star(\boldsymbol{B},0,r_1,r_2)^\top \boldsymbol{\Xi}^\star(\boldsymbol{B},0,r_1,r_2)^{-1} \boldsymbol{D}^\star(\boldsymbol{B},0,r_1,r_2) \\ &\quad + \sup_{(s_1,s_2)\in(\mathcal{G}_\varepsilon\times[0,1])\cap\Omega(\varepsilon)} \boldsymbol{D}^\star(\boldsymbol{B},s_1,s_2,1)^\top \boldsymbol{\Xi}^\star(\boldsymbol{B},s_1,s_2,1)^{-1} \boldsymbol{D}^\star(\boldsymbol{B},s_1,s_2,1); \end{aligned}$$

and (ii) under the alternative (10) with $\min_{0\le i\le M^\star} |k_{i+1}^\star - k_i^\star|/N > 3\varepsilon/2$ *and* $\boldsymbol{\theta}_{k_i^\star+1} - \boldsymbol{\theta}_{k_i^\star} = N^{-1/2} L_i^\star$ *for some* $L_i^\star \neq 0$, $i = 1,\ldots,M^\star$, *we have* $G_N^\diamond \to \infty$ *in probability if* $\min_{1\le i\le M^\star} |L_i^\star| \to \infty$.

Therefore, the cut-off values can be obtained by simulating the distribution of $G^\diamond(\boldsymbol{B})$, which

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is often achieved in practice by simulating the test statistic $G_N^\diamond$ for lengthy Gaussian white noises. Since a large number of simulated replications are needed to accurately estimate the cut-off values, reducing the computational burden needed for computing the test statistic for each realization can be of significant importance. It can be seen from our simulation results in Section 4.2 that the above grid approximation scheme leads to tests with reasonably good performance in terms of both the size and power.

4.2 Simulation Results

We shall here carry out Monte Carlo simulations to examine the finite-sample performance of the proposed hypothesis testing procedure and make a comparison with the test of Shao and Zhang (2010). Due to the nonstandard limiting null distribution, critical values of both tests are obtained via simulation based on $n = 2{,}500$ and 10,000 replications; see also Shao and Zhang (2010). Let w_i, $i \in \mathbb{Z}$, be independent standard normal random variables, and (e_i) be an autoregressive process satisfying the recursion $e_i = \rho e_{i-1} + w_i$. Let $\mathbb{1}_{\{\cdot\}}$ denote the indicator function, we consider the model

$$X_i = \mu_i + e_i, \quad i = 1, \ldots, n, \tag{18}$$

with the following change-point alternatives:

(M1) (one change-point alternative): $\mu_i = d\mathbb{1}_{\{i/n>2/3\}}$;

(M2) (two change-point alternative): $\mu_i = d\mathbb{1}_{\{2/3\geq i/n>1/3\}} - d\mathbb{1}_{\{i/n>2/3\}}$;

(M3) (three change-point alternative): $\mu_i = d\mathbb{1}_{\{i/n>1/4\}} - d\mathbb{1}_{\{3/4\geq i/n>1/2\}}$,

where $d \in \mathbb{R}$ controls the amount of deviation from the null. For (M1)–(M3), the case with $d = 0$ corresponds to the null hypothesis of no change point. The autoregressive process considered here belongs to the class of weakly dependent processes, whose invariance principles have been studied by Wu (2007), Shao (2010b) and Volgushev and Shao (2014) among many others; see also Shao and Zhang (2010), Berkes et al. (2014) and Shao (2015) for additional references. Let $n = 500$ and $\rho \in \{\pm 0.3, \pm 0.6, \pm 0.8\}$, we consider testing for change points in the mean and median of the process (18). For this, we consider the proposed new self-normalized test, namely the unsupervised contrast-based self-normalized (UCbSN) test, and compare it with the recent self-normalized test of Shao and Zhang (2010), which we will abbreviate as SZ10 hereafter; see also Shao and Zhang (2010) for advantages of using self-normalization in change-point testing. Note that the SZ10 test

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is supervised and requires a prespecified value for the total number of change points, which however is typically unknown in practice especially when one is at the stage of seeking a statistical test to determine their existence. Therefore, we shall here consider scenarios when this piece of prior information is correctly specified and when it is misspecified. In particular, we consider SZ10 test statistics constructed under one, two and three hypothesized change points, which we denote by $SZ10_1$, $SZ10_2$ and $SZ10_3$ respectively. For the proposed unsupervised test, it does not require this piece of prior information and the grid approximation scheme discussed in Section 4.1 is implemented to facilitate the computation. For the choice of the trimming parameter ε, Zhou and Shao (2013) and Huang et al. (2015) conducted a series of numerical analyses and argued that the rule of thumb choice of $\varepsilon = 0.1$ seems to yield satisfactory performance, which we shall use in our numerical analysis. Note that the effect of trimming, unlike the effect of smoothing as in nonparametric estimation, is accounted for in the limiting distribution and its approximation; see also the discussion in Zhou and Shao (2013). For each configuration, we generate 2,500 realizations, and the results are summarized in Sections 4.2.1 and 4.2.2 for comparisons under the null and under the alternative respectively. Additional simulation results can be found in the online supplementary material.

4.2.1 Comparison: Empirical Acceptance Rate

We first examine the performance under the null, and Table 1 provides the empirical acceptance rates for 90% and 95% nominal levels. For the SZ10 test, it is supervised and it can be seen from Table 1 that using a larger prespecified number of change points generally leads to a larger size distortion for their test. In addition, size distortions of $SZ10_2$ and $SZ10_3$, multiple change-point SZ10 tests hereby, can deteriorate quickly from the mean case to the median case. In particular, the empirical acceptance rate of the $SZ10_3$ test is 0.572 for the median case even when $\rho = 0.3$, which is far away from its nominal 95% level. In comparison, the proposed test is unsupervised and does not require prespecifying the number of change points. Its performance in size is comparable to the best of multiple change-point SZ10 tests in the mean case, and can largely improve over both $SZ10_2$ and $SZ10_3$ in the median case. Therefore, besides being unsupervised, the proposed test also seems to deliver more reasonable performance in size than its multiple change-point SZ10 competitors. The $SZ10_1$ test, though being designed only for cases with a single change point, seems to have the best performance in size due to its simpler form. However, it can be seen from Section 4.2.2 that the single change-point $SZ10_1$ test can suffer from a serious power issue in the

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presence of multiple change points.

4.2.2 Comparison: Empirical Power

We shall now consider (M1)–(M3) and examine the performance under the alternative. Tables 2–4 provide the size-adjusted empirical powers under (M1)–(M3) respectively for $\rho \in \{0.3, 0.6, 0.8\}$, and results with additional choices of ρ can be found in the online supplementary material. For the SZ10 test, it requires prespecifying the number of change points and it can be seen from Tables 2–4 that, depending on whether this prior information is correctly specified or not, the power performance of their test can be largely affected. For example, when there is only a single change point as in (M1), then prespecifying two or more change points can lead to serious power losses for the SZ10 test, as empirical powers of $SZ10_2$ and $SZ10_3$ in Table 2 are in general much lower than those of $SZ10_1$ and the proposed UCbSN. In this case, the single change-point $SZ10_1$ test performs the best, which also outperforms the proposed test though the difference seems to be subtle when ρ is not too large. However, when there are multiple change points as in (M2) and (M3), then the single change-point $SZ10_1$ test can suffer from serious power losses as can be seen from Tables 3 and 4. In contrast, the proposed UCbSN test does not require prespecifying the number of change points and continues to deliver a reasonably good power performance for cases with multiple change points. In particular, it can be seen from Tables 3 and 4 that empirical powers of the proposed test are mostly comparable to the oracle SZ10 test in the mean case and can outperform all the SZ10 tests considered in the median case. The oracle SZ10 test is defined as the SZ10 test where the number of change points is assumed to be known and correctly prespecified, which corresponds to $SZ10_2$ and $SZ10_3$ under (M2) and (M3) respectively. The oracle SZ10 test represents the power upper bound of SZ10 tests but is typically not obtainable in practice due to the unknown number of change points. Therefore, in terms of the power performance, the proposed test seems to be more reliable in situations where the number of change points is unknown.

4.3 A Real Data Analysis

We shall here apply the results to a climate science data for assessing the temporal stability or stationarity of tropical cyclones. It has been argued and believed in the climate science literature that tropical cyclone winds should increase with increasing ocean temperature, one of the major consequences of global warming. As a result, researchers have been seeking empirical evidences on changes in tropical cyclone winds; see for example Elsner et al. (2008), Zhou (2010), Zhang

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and Wu (2011) and references therein. Analyses in the aforementioned papers require consistent estimation of the nuisance long-run variance, and we shall here use self-normalization to alleviate the difficult issue of estimating the long-run variance under the change-point alternative; see Shao (2010b) and Shao and Zhang (2010) for advantages of using self-normalization. A time series plot of the data is given in Figure 1, which contains satellite-derived lifetime-maximum wind speeds of 2098 tropical cyclones over the globe during 1981–2006, and we refer to Elsner et al. (2008) for a more detailed description. The results are summarized in Table 5, and a comparison with the supervised test of Shao and Zhang (2010) is also given. It can be seen from Table 5 that, at 5% significance level, the proposed unsupervised test finds statistical evidence against the hypotheses of mean constancy and variance constancy, indicating changes in tropical cyclone winds. As a comparison, results from the test of Shao and Zhang (2010) seem less conclusive and depend on what number of change points one would like to prespecify for their test. Since the number of change points is typically unknown in practice especially when one is at the stage of seeking statistical tests to determine their existence, the proposed unsupervised test seems to provide a useful alternative for self-normalized change-point detection.

5 Conclusion

Since the influential work of Shao (2010b), self-normalization has experienced deep and exciting developments in various statistical inference problems, and a recent review is given by Shao (2015). However, the majority of existing results focused on making inference of quantities associated with a known stationary time series, and as commented by Shao and Zhang (2010), its adaptation to change-point testing problems can be nontrivial as direct implementation can lead to inconsistent tests. The aforementioned paper considered the situation with a single change-point alternative and proposed a consistent self-normalized test. Its extension to multiple change-point alternatives was also given but requires prespecifying the actual number of change points under the alternative and is thus supervised. Since the actual number of change points is typically unknown especially when one is at the stage of seeking a statistical test to determine their existence, it seems desirable if we can have an unsupervised counterpart as considered in the current paper that can be used for situations when there is no or ambiguous prior knowledge about the number of change points or when doubts are casted on this prior knowledge. As can be seen from our simulation study in Section 4.2, the supervised test of Shao and Zhang (2010) can suffer from serious power losses when the prespecified value on the number of change points is misspecified,

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and in certain situations the power of their test can even decrease to zero as the alternative deviates from the null. In contrast, the proposed unsupervised test does not require prespecifying the number of change points and seems to perform reasonably well for situations with different numbers of change points. In addition, we propose a new approach, called the contrast-based method, for generalizing self-normalized statistics to quantities other than the mean. Unlike the conventional approach which was mainly developed for constructing confidence intervals for quantities associated with a stationary time series, the proposed contrast-based approach is specifically tailored for change-point testing problems and can lead to tests with better power performance; see also the discussion in Section 3. Another useful feature of the proposed contrast-based approach is that it allows one to study the asymptotic power behavior of self-normalized tests under a more desirable and interpretable deterministic condition as in Theorem 3.2 (ii); see condition (16) for a comparison. This also complements the result of Shao and Zhang (2010) who only studied the asymptotic power of their test for the mean case.

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Appendix

We shall here provide the technical proofs for results in Sections 2–4.

Proof. (Theorems 2.1 and 3.1) We shall first provide the proof for the mean case, namely Theorem 2.1. For this, note that the partial sum $S_{j,k} = S_{1,k} - S_{1,j-1} = (k-j+1)\mu + \sqrt{n}[S_n(k/n) - S_n\{(j-1)/n\}]$, $1 \le j \le k \le n$, we have

$$\begin{aligned} & \sqrt{\lfloor nr_2 \rfloor / n} D_{n,f}(1, \lfloor nr_1 \rfloor, \lfloor nr_2 \rfloor) - \sqrt{r_2} D(S_n, 0, r_1, r_2) \\ = \; & \{S_n(\lfloor nr_1 \rfloor / n) - (\lfloor nr_1 \rfloor / \lfloor nr_2 \rfloor) S_n(\lfloor nr_2 \rfloor / n)\} - \{S_n(r_1) - (r_1/r_2) S_n(r_2)\} \\ = \; & \{(r_1/r_2) - (\lfloor nr_1 \rfloor / \lfloor nr_2 \rfloor)\} S_n(r_2). \end{aligned}$$

Under condition (IP), both $\sup_{t\in[0,1]} |S_n(t)|$ and $\sup_{(r_1,r_2)\in\Omega(\varepsilon)} |D_{n,f}(1, \lfloor nr_1 \rfloor, \lfloor nr_2 \rfloor)|$ are of order $O_p(1)$.

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Since $\sup_{(r_1,r_2)\in\Omega(\varepsilon)} |(r_1/r_2) - (\lfloor nr_1\rfloor/\lfloor nr_2\rfloor)| \leq 2/\{n(2\varepsilon - 1/n)^2\}$, we have

$$\sup_{(r_1,r_2)\in\Omega(\varepsilon)} |\sqrt{\lfloor nr_2\rfloor/n} D_{n,f}(1, \lfloor nr_1\rfloor, \lfloor nr_2\rfloor) - \sqrt{r_2} D(S_n, 0, r_1, r_2)| = O_p(n^{-1}),$$

and thus

$$\sup_{(r_1,r_2)\in\Omega(\varepsilon)} |D_{n,f}(1, \lfloor nr_1\rfloor, \lfloor nr_2\rfloor) - D(S_n, 0, r_1, r_2)| = O_p(n^{-1}).$$

Note that $\sup_{(r_1,r_2)\in\Omega(\varepsilon)} |D_{n,f}(1, \lfloor nr_1\rfloor, \lfloor nr_2\rfloor)| = \max_{(l_1,l_2)\in\Omega_n(\varepsilon)} |D_{n,f}(1, l_1, l_2)|$ holds for all large n, by a similar argument and properties of the Brownian motion, it can be shown that

$$\left| T_n - \left\{ \sup_{(r_1,r_2)\in\Omega(\varepsilon)} \frac{D(S_n, 0, r_1, r_2)^2}{\Xi(S_n, 0, r_1, r_2)} + \sup_{(s_1,s_2)\in\Omega(\varepsilon)} \frac{D(S_n, s_1, s_2, 1)^2}{\Xi(S_n, s_1, s_2, 1)} \right\} \right| = o_p(1),$$

and (i) follows by the continuous mapping theorem. For (ii), under the alternative (2) with $E(X_{k_{i+1}}) - E(X_{k_i}) = n^{-1/2} L_i$, $i = 1, \ldots, M$, we have

$$\begin{aligned} D_{n,f}(1, k_1, k_2) &= \frac{1}{\sqrt{k_2}} \left[\sum_{i=1}^{k_1} \{X_i - E(X_i)\} - \frac{k_1}{k_2} \sum_{j=1}^{k_2} \{X_j - E(X_j)\} \right] - \frac{k_1(k_2 - k_1)}{k_2 \sqrt{nk_2}} L_1 \\ &:= D_{n,f,e}(1, k_1, k_2) - \frac{k_1(k_2 - k_1)}{k_2 \sqrt{nk_2}} L_1. \end{aligned}$$

By the argument of (i) and the condition that $\min_{0\leq i\leq M} |k_{i+1} - k_i|/n > \varepsilon$, one has

$$\frac{D_{n,f,e}(1, k_1, k_2)^2}{\Xi_{n,f}(1, k_1, k_2)} = O_p(1).$$

Note that $k_1(k_2 - k_1)/(k_2 \sqrt{nk_2}) \geq \varepsilon^2$, the result follows by $\min_{1\leq i\leq M} |L_i| \to \infty$. A similar argument can be used to prove Theorem 3.1, and we shall here only highlight the key differences. In particular, under the general setting as framed in Section 3, $\hat{\boldsymbol{\theta}}_{j,k}$, $j \leq k$, can no longer be expressed as a linear combination of the recursive statistics $\hat{\boldsymbol{\theta}}_{1,i}$, $i = 1, \ldots, n$. For this, we let

$$\begin{aligned} \boldsymbol{\Delta}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}}, r, s, t, u) &= \frac{1}{\sqrt{N}} \{(\lfloor Ns\rfloor - \lfloor Nr\rfloor \vee 1 + 1) \boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor Nr\rfloor\vee 1, \lfloor Ns\rfloor}) \\ &\quad - (u - t)^{-1}(s - r)(\lfloor Nu\rfloor - \lfloor Nt\rfloor \vee 1 + 1) \boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor Nt\rfloor\vee 1, \lfloor Nu\rfloor})\}, \end{aligned}$$

and define

$$\boldsymbol{D}^{\star}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}}, t_1, t_2, t_3) = \frac{1}{\sqrt{t_3 - t_1}} \boldsymbol{\Delta}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}}, t_1, t_2, t_1, t_3).$$

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Note that

$$\begin{aligned}&\sqrt{\lfloor Nr_2\rfloor/N}\,\boldsymbol{D}^{\star}_{N,f}(1,\lfloor Nr_1\rfloor,\lfloor Nr_2\rfloor)-\sqrt{r_2}\,\boldsymbol{D}^{\star}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}},0,r_1,r_2)\\ =\;&\left(\frac{r_1}{r_2}-\frac{\lfloor Nr_1\rfloor}{\lfloor Nr_2\rfloor}\right)\frac{\lfloor Nr_2\rfloor}{\sqrt{N}}\boldsymbol{Q}(\hat{\boldsymbol{F}}_{1,\lfloor Nr_2\rfloor}),\end{aligned}$$

and thus by condition (IP*), one has

$$\sup_{(r_1,r_2)\in\Omega(\varepsilon)}|\sqrt{\lfloor Nr_2\rfloor/N}\,\boldsymbol{D}^{\star}_{N,f}(1,\lfloor Nr_1\rfloor,\lfloor Nr_2\rfloor)-\sqrt{r_2}\,\boldsymbol{D}^{\star}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}},0,r_1,r_2)|=O_p(N^{-1}).$$

The remaining of the proof follows by a similar argument as in Theorem 2.1. □

Proof. (Proposition 3.1) In the case of the mean, we have $\hat{\theta}_{j,k}=S_{j,k}/(k-j+1)=(k-j+1)^{-1}\sum_{i=j}^{k}\{X_i-E(X_i)\}+(k-j+1)^{-1}\sum_{i=j}^{k}E(X_i)$. Let $S^{\circ}_{j,k}=S_{j,k}-E(S_{j,k})$ denote the centered partial sum and $d_i^\star=E(X_{k_i^\star+1})-E(X_{k_i^\star})$ represent the magnitude of change, then

$$\begin{aligned}\sqrt{N}|\hat{\theta}_{k^\star_{i-1}+1,k^\star_i}-\hat{\theta}_{k^\star_{i-1}+1,k^\star_{i+1}}| &= \sqrt{N}\left|\frac{S_{k^\star_{i-1}+1,k^\star_i}}{k^\star_i-k^\star_{i-1}}-\frac{S_{k^\star_{i-1}+1,k^\star_{i+1}}}{k^\star_{i+1}-k^\star_{i-1}}\right|\\ &= \sqrt{N}\left|\frac{S^{\circ}_{k^\star_{i-1}+1,k^\star_i}}{k^\star_i-k^\star_{i-1}}-\frac{S^{\circ}_{k^\star_{i-1}+1,k^\star_{i+1}}}{k^\star_{i+1}-k^\star_{i-1}}-\frac{k^\star_{i+1}-k^\star_i}{k^\star_{i+1}-k^\star_{i-1}}d^\star\right|\\ &\geq \sqrt{N}\left(\frac{k^\star_{i+1}-k^\star_i}{k^\star_{i+1}-k^\star_{i-1}}|d^\star|-\left|\frac{S^{\circ}_{k^\star_{i-1}+1,k^\star_i}}{k^\star_i-k^\star_{i-1}}-\frac{S^{\circ}_{k^\star_{i-1}+1,k^\star_{i+1}}}{k^\star_{i+1}-k^\star_{i-1}}\right|\right).\end{aligned}$$

By condition (IP) and the proof of Theorem 2.1,

$$\sqrt{N}\left|\frac{S^{\circ}_{k^\star_{i-1}+1,k^\star_i}}{k^\star_i-k^\star_{i-1}}-\frac{S^{\circ}_{k^\star_{i-1}+1,k^\star_{i+1}}}{k^\star_{i+1}-k^\star_{i-1}}\right|=O_p(1),$$

and the result follows. □

Proof. (Theorem 3.2) Let

$$\begin{aligned}\boldsymbol{D}^{\diamond}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}},t_1,t_2,t_3) &= \frac{1}{\sqrt{N(t_3-t_1)^3}}\{(t_3-t_2)(\lfloor Nt_2\rfloor-\lfloor Nt_1\rfloor\vee 1+1)\boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor Nt_1\rfloor\vee 1,\lfloor Nt_2\rfloor})\\ &\quad-(t_2-t_1)(\lfloor Nt_3\rfloor-\lfloor Nt_2\rfloor)\boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor Nt_2\rfloor+1,\lfloor Nt_3\rfloor})\},\end{aligned}$$

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then one can write

$$\begin{aligned}
&\boldsymbol{D}^{\diamond}_{N,f}(1,\lfloor Nr_1\rfloor,\lfloor Nr_2\rfloor) - \boldsymbol{D}^{\diamond}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}},0,r_1,r_2)\\
=\ &\left\{\frac{(\lfloor Nr_2\rfloor-\lfloor Nr_1\rfloor)/N}{(\lfloor Nr_2\rfloor/N)^{3/2}} - \frac{r_2-r_1}{r_2^{3/2}}\right\}\frac{\lfloor Nr_1\rfloor}{\sqrt{N}}\boldsymbol{Q}(\hat{\boldsymbol{F}}_{1,\lfloor Nr_1\rfloor})\\
&-\left(\frac{\lfloor Nr_1\rfloor/N}{(\lfloor Nr_2\rfloor/N)^{3/2}} - \frac{r_1}{r_2^{3/2}}\right)\frac{\lfloor Nr_2\rfloor-\lfloor Nr_1\rfloor}{\sqrt{N}}\boldsymbol{Q}(\hat{\boldsymbol{F}}_{\lfloor Nr_1\rfloor+1,\lfloor Nr_2\rfloor}).
\end{aligned}$$

Note that for any $t\in[0,1]$, $t-1/N<\lfloor Nt\rfloor/N\le t$, we have by condition (IP*),

$$\sup_{(r_1,r_2)\in\Omega(\varepsilon)}|\boldsymbol{D}^{\diamond}_{N,f}(1,\lfloor Nr_1\rfloor,\lfloor Nr_2\rfloor) - \boldsymbol{D}^{\diamond}_{\boldsymbol{Q},N}(\hat{\boldsymbol{F}},0,r_1,r_2)| = O_p(N^{-1}),$$

and (i) follows by the continuous mapping theorem and a similar argument as in Theorem 2.1. For (ii), under the alternative (10) with $\boldsymbol{\theta}_{k^\star_{i+1}}-\boldsymbol{\theta}_{k^\star_i}=N^{-1/2}L^\star_i$, $i=1,\ldots,M^\star$, we have

$$\boldsymbol{D}^{\diamond}_{N,f}(1,k^\star_1,k^\star_2) = \frac{k^\star_1(k^\star_2-k^\star_1)}{(k^\star_2)^{3/2}}\{(\hat{\boldsymbol{\theta}}_{1,k^\star_1}-\boldsymbol{\theta}_{k^\star_1})-(\hat{\boldsymbol{\theta}}_{k^\star_1+1,k^\star_2}-\boldsymbol{\theta}_{k^\star_1+1})\} - \frac{k^\star_1(k^\star_2-k^\star_1)}{k^\star_2\sqrt{Nk^\star_2}}\boldsymbol{L}^\star_1.$$

Under the change-point alternative, we have $\boldsymbol{\theta}_1=\cdots=\boldsymbol{\theta}_{k^\star_1}$ and $\boldsymbol{\theta}_{k^\star_1+1}=\cdots=\boldsymbol{\theta}_{k^\star_2}$. Hence, by condition (IP*) and the argument of (i), one can obtain that

$$\frac{k^\star_1(k^\star_2-k^\star_1)}{(k^\star_2)^{3/2}}\{(\hat{\boldsymbol{\theta}}_{1,k^\star_1}-\boldsymbol{\theta}_{k^\star_1})-(\hat{\boldsymbol{\theta}}_{k^\star_1+1,k^\star_2}-\boldsymbol{\theta}_{k^\star_1+1})\} = O_p(1).$$

The result then follows by noticing that $\min_{1\le i\le M^\star}|L^\star_i|\to\infty$. □

Proof. (Proposition 4.1) Note that $(l_1,l_2)\in\mathcal{G}_{\varepsilon,N,f}$ if and only if $(l_1,l_2)=(\lfloor Nr_1\rfloor,\lfloor Nr_2\rfloor)$ for some $(r_1,r_2)\in([0,1]\times\mathcal{G}_\varepsilon)\cap\Omega(\varepsilon)$, the first claim follows by the argument of Theorem 3.2 and the continuous mapping theorem. We shall now provide the proof for (ii). For this, note that for any $l_2\in(k^\star_1,k^\star_2]$,

$$\boldsymbol{D}^{\diamond}_{n,f}(1,k^\star_1,l_2) = \frac{k^\star_1(l_2-k^\star_1)}{l_2^{3/2}}\{(\hat{\boldsymbol{\theta}}_{1,k^\star_1}-\boldsymbol{\theta}_{k^\star_1})-(\hat{\boldsymbol{\theta}}_{k^\star_1+1,l_2}-\boldsymbol{\theta}_{k^\star_1+1})\} - \frac{k^\star_1(l_2-k^\star_1)}{l_2\sqrt{Nl_2}}\boldsymbol{L}^\star_1,$$

then by condition (IP*), it suffices to prove that there exists a sequence $l_2(N)\in(k^\star_1,k^\star_2]$ such that $\{k^\star_1,l_2(N)\}\in\mathcal{G}_{\varepsilon,N,f}$ and $\{l_2(N)-k^\star_1\}/l_2(N)$ is bounded away from zero for all large N. For each $N=1,2,\ldots$, by the construction of $\mathcal{G}_\varepsilon$, there exists an element, denoted by a_N, such that $a_N\in\mathcal{G}_\varepsilon\cap(k^\star_2/N-\varepsilon/2,k^\star_2/N]$. Note that $(Na_N-k^\star_1)/(Na_N)\ge(k^\star_2)^{-1}(N\varepsilon)\ge\varepsilon$ and $(Na_N-\lfloor Na_N\rfloor)/N\le N^{-1}$, the result follows by letting $l_2(N)=\lfloor Na_N\rfloor\vee 1$. □

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	$SZ10_1$		$SZ10_2$		$SZ10_3$		UCbSN	
ρ	90%	95%	90%	95%	90%	95%	90%	95%
			Results for the mean					
0.3	0.896	0.950	0.882	0.942	0.890	0.938	0.890	0.950
0.6	0.897	0.944	0.868	0.929	0.854	0.913	0.862	0.932
0.8	0.889	0.936	0.795	0.881	0.724	0.807	0.784	0.866
-0.3	0.892	0.950	0.910	0.955	0.933	0.968	0.910	0.958
-0.6	0.909	0.958	0.941	0.972	0.963	0.983	0.932	0.968
-0.8	0.920	0.963	0.958	0.981	0.984	0.994	0.953	0.976
			Results for the median					
0.3	0.867	0.920	0.743	0.825	0.456	0.572	0.835	0.904
0.6	0.880	0.936	0.723	0.817	0.493	0.602	0.805	0.891
0.8	0.872	0.927	0.697	0.792	0.434	0.534	0.765	0.845
-0.3	0.872	0.926	0.682	0.778	0.348	0.458	0.825	0.900
-0.6	0.869	0.922	0.655	0.740	0.269	0.357	0.842	0.906
-0.8	0.859	0.912	0.635	0.727	0.213	0.297	0.842	0.910

Table 1: Empirical acceptance rates for testing change points in the mean and median of (18) under different dependence strengths. For both the proposed UCbSN test and the SZ10 tests of Shao and Zhang (2010), results with 90% and 95% nominal levels are reported.

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		Mean				*Median*			
ρ	d	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN
0.3	0.2	0.198	0.110	0.074	0.160	0.122	0.070	0.052	0.109
	0.4	0.544	0.292	0.170	0.486	0.334	0.154	0.099	0.330
	0.6	0.845	0.598	0.405	0.833	0.613	0.328	0.164	0.644
	0.8	0.962	0.856	0.657	0.970	0.812	0.532	0.302	0.868
	1.0	0.991	0.971	0.872	0.996	0.935	0.741	0.478	0.972
	2.0	1.000	1.000	1.000	1.000	0.999	0.998	0.973	1.000
	3.0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.6	0.2	0.098	0.068	0.056	0.084	0.085	0.052	0.066	0.057
	0.4	0.216	0.108	0.074	0.181	0.163	0.068	0.070	0.121
	0.6	0.410	0.197	0.125	0.354	0.319	0.124	0.102	0.262
	0.8	0.623	0.354	0.216	0.593	0.493	0.199	0.161	0.444
	1.0	0.794	0.525	0.321	0.781	0.664	0.326	0.222	0.637
	2.0	0.998	0.984	0.936	0.998	0.979	0.882	0.743	0.991
	3.0	1.000	1.000	0.998	1.000	0.997	0.996	0.972	1.000
0.8	0.2	0.062	0.048	0.042	0.050	0.068	0.054	0.042	0.065
	0.4	0.100	0.060	0.055	0.061	0.079	0.059	0.055	0.074
	0.6	0.143	0.075	0.059	0.093	0.116	0.072	0.054	0.100
	0.8	0.210	0.086	0.065	0.127	0.175	0.071	0.057	0.129
	1.0	0.306	0.118	0.086	0.198	0.227	0.094	0.071	0.198
	2.0	0.776	0.434	0.245	0.682	0.634	0.325	0.200	0.623
	3.0	0.968	0.819	0.588	0.950	0.870	0.654	0.458	0.913

Table 2: Size-adjusted empirical powers for testing change points in the mean and median of (18) under (M1) with different dependence strengths. For both the proposed UCbSN test and the SZ10 tests of Shao and Zhang (2010), the significance level $\alpha = 0.05$ is used.

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		Mean				*Median*			
ρ	d	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN
0.3	0.2	0.132	0.316	0.243	0.368	0.102	0.167	0.126	0.219
	0.4	0.208	0.850	0.776	0.878	0.151	0.522	0.340	0.700
	0.6	0.166	0.992	0.988	0.996	0.144	0.840	0.675	0.953
	0.8	0.141	1.000	1.000	1.000	0.100	0.971	0.904	0.996
	1.0	0.110	1.000	1.000	1.000	0.057	0.997	0.981	1.000
	2.0	0.008	1.000	1.000	1.000	0.000	1.000	1.000	1.000
	3.0	0.000	1.000	1.000	1.000	0.000	1.000	1.000	1.000
0.6	0.2	0.086	0.115	0.098	0.129	0.074	0.095	0.081	0.117
	0.4	0.154	0.346	0.251	0.382	0.137	0.212	0.160	0.316
	0.6	0.178	0.682	0.542	0.707	0.157	0.412	0.321	0.565
	0.8	0.195	0.902	0.824	0.912	0.164	0.677	0.556	0.830
	1.0	0.166	0.975	0.954	0.984	0.130	0.839	0.739	0.940
	2.0	0.058	1.000	1.000	1.000	0.015	0.998	0.999	1.000
	3.0	0.014	1.000	1.000	1.000	0.000	1.000	1.000	1.000
0.8	0.2	0.056	0.058	0.051	0.056	0.051	0.064	0.058	0.058
	0.4	0.086	0.120	0.084	0.112	0.067	0.083	0.070	0.097
	0.6	0.124	0.177	0.124	0.195	0.098	0.121	0.091	0.161
	0.8	0.153	0.321	0.206	0.330	0.127	0.196	0.143	0.284
	1.0	0.184	0.491	0.308	0.508	0.148	0.309	0.218	0.422
	2.0	0.159	0.965	0.892	0.968	0.094	0.841	0.760	0.910
	3.0	0.080	0.998	0.998	0.999	0.020	0.982	0.984	0.998

Table 3: Size-adjusted empirical powers for testing change points in the mean and median of (18) under (M2) with different dependence strengths. For both the proposed UCbSN test and the SZ10 tests of Shao and Zhang (2010), the significance level $\alpha = 0.05$ is used.

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		Mean				*Median*			
ρ	d	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN
0.3	0.2	0.051	0.111	0.119	0.120	0.053	0.084	0.075	0.092
	0.4	0.040	0.225	0.346	0.284	0.046	0.142	0.148	0.201
	0.6	0.023	0.526	0.732	0.662	0.036	0.250	0.284	0.444
	0.8	0.006	0.792	0.937	0.905	0.021	0.455	0.470	0.741
	1.0	0.001	0.941	0.994	0.987	0.010	0.625	0.696	0.916
	2.0	0.000	1.000	1.000	1.000	0.000	0.991	0.998	1.000
	3.0	0.000	1.000	1.000	1.000	0.000	1.000	1.000	1.000
0.6	0.2	0.048	0.070	0.070	0.064	0.052	0.053	0.053	0.062
	0.4	0.058	0.118	0.128	0.110	0.050	0.074	0.074	0.082
	0.6	0.042	0.184	0.260	0.201	0.056	0.115	0.138	0.152
	0.8	0.034	0.306	0.430	0.379	0.047	0.183	0.218	0.287
	1.0	0.020	0.455	0.636	0.566	0.036	0.256	0.320	0.430
	2.0	0.002	0.968	0.996	0.991	0.006	0.780	0.882	0.968
	3.0	0.000	1.000	1.000	1.000	0.000	0.979	0.997	0.999
0.8	0.2	0.050	0.058	0.052	0.054	0.050	0.054	0.051	0.050
	0.4	0.047	0.059	0.061	0.055	0.044	0.055	0.065	0.064
	0.6	0.048	0.074	0.084	0.074	0.044	0.062	0.068	0.070
	0.8	0.056	0.086	0.109	0.100	0.055	0.087	0.086	0.093
	1.0	0.048	0.102	0.140	0.118	0.058	0.094	0.104	0.121
	2.0	0.027	0.366	0.528	0.507	0.027	0.265	0.316	0.407
	3.0	0.006	0.730	0.891	0.883	0.010	0.558	0.668	0.779

Table 4: Size-adjusted empirical powers for testing change points in the mean and median of (18) under (M3) with different dependence strengths. For both the proposed UCbSN test and the SZ10 tests of Shao and Zhang (2010), the significance level $\alpha = 0.05$ is used.

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	Mean				*Variance*			
	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN	$SZ10_1$	$SZ10_2$	$SZ10_3$	UCbSN
Test statistic	35.27	184.42	225.26	198.67	97.42	133.65	135.47	210.84
p-value	0.070	0.012	0.061	0.011	0.003	0.052	0.329	0.008

Table 5: Test statistics and their *p*-values for the tropical cyclone data.

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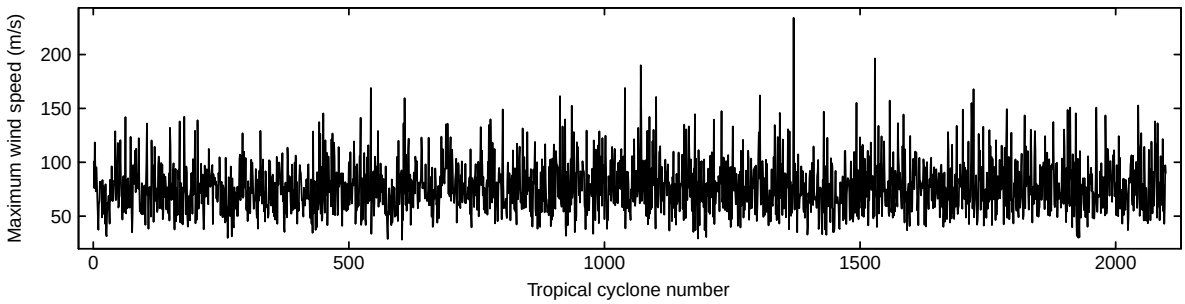

Figure 1: Satellite-derived lifetime-maximum wind speeds of tropical cyclones during 1981–2006.

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